

Quantitative Reasoning

GQR = 101

Unit # 05

Area, Perimeter, and Volume

Question # 01

One of the sides and the corresponding height of a parallelogram are 5cm and 3cm respectively. Find the area of the parallelogram.

$$\text{Area} = \text{Base} \times \text{Height}$$

Given:-

- Base (one side) = 5cm
- Corresponding height = 3cm

Plugging in the values:-

$$\begin{aligned}\text{Area} &= 5\text{cm} \times 3\text{cm} \\ &= (5)(3)\text{cm}^2 \\ &= 15\text{cm}^2\end{aligned}$$

Question # 02

In a triangle, if one side is 7cm and its corresponding height is 4cm, find its area.

$$\text{Area} = \frac{1}{2} (\text{Base} \times \text{height})$$

Given:-

- Base (one side) = 7cm

- Corresponding height = 4cm

Plugging in the values:-

$$\text{Area} = \frac{1}{2} (7\text{cm} \times 4\text{cm})$$

$$= \frac{1}{2} (28) \text{cm}^2$$

$$= \frac{1}{2} \times 28 \text{cm}^2$$

$$= 14 \text{cm}^2$$

Question # 03

If area of $\triangle ABC$ is 49cm^2 and its height AD is 7cm find BC.

$$\text{Area of a triangle} = \frac{1}{2} (\text{Base} \times \text{Height})$$

$$49 \text{ cm}^2 = \frac{1}{2} BC \times 7 \text{ cm}$$

Given:

- Area of $\triangle ABC = 49 \text{ cm}^2$
- Height $AD = 7 \text{ cm}$

Given:

$$BC = (2 \times \text{Area}) / \text{height}$$

Plugging in the values:

$$BC = (2 \times 49 \text{ cm}^2) / 7$$

$$BC = \frac{98 \text{ cm}^2}{7}$$

$$= 14$$

Question # 04

If area of a parallelogram is 17.6 cm^2 and its base is 4 cm , find its corresponding height.

Area of a parallelogram = Base $\times$ height

Given:

- Base (one side) = 4 cm
- Area of a parallelogram = 17.6 cm^2

Plugging in the values:

$$\text{Height} = \frac{17.6 \text{ cm}^2}{4 \text{ cm}}$$

$$= 4.4 \text{ cm}$$

Question # 05

Find the breadth of a rectangular plot if its area is 440 cm^2 and length is 22 cm . Also find its perimeter.

Area of a rectangle = length $\times$ Breadth

- Area of a rectangle = 440 cm^2
- Length = 22 cm

Plugging in the values:-

$$\begin{aligned}\text{Breadth} &= \frac{440 \text{ cm}^2}{22 \text{ cm}} \\ &= 20 \text{ cm}\end{aligned}$$

$$\begin{aligned}\text{Perimeter of a rectangle} &= 2(\text{Length} + \text{Breadth}) \\ &= 2(22 \text{ cm} + 20 \text{ cm}) \\ &= 2(42) \text{ cm} \\ &= 84 \text{ cm}\end{aligned}$$

Question # 06

The perimeter of a rectangular sheet is 100 cm . If the length is 35 cm , find its breadth.

$$\text{Perimeter of a rectangle} = 2 \times (\text{Length} + \text{Breadth})$$

- rectangle sheet = 100cm
- length = 35

Plugging in the values:-
 $100\text{cm} = 2(35\text{cm} + \text{Breadth})$

$$\frac{100\text{cm}}{2} = 35\text{cm} + \text{Breadth}$$

$$50\text{cm} = 35\text{cm} + \text{Breadth}$$

$$\text{Breadth} = 50\text{cm} - 35\text{cm} = \text{Breadth} =$$

$$\text{Breadth} = 15\text{cm}$$

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